MediScan

Visual Medicine Finder — A Multi-Model Deep Learning Web Application

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1. Introduction

MediScan is a deep learning-powered web application designed for Indian users — particularly the elderly, illiterate, and accessibility-constrained population. A user simply photographs a medicine strip or a doctor's handwritten prescription. The system automatically reads the medicine name, retrieves cheaper generic alternatives with the same salt composition, displays side effect severity levels, checks for dangerous drug combinations, and detects tampered or counterfeit packaging. No typing is required at any stage. Five independent deep learning models work together in a single Django-based pipeline, addressing medicine cost, patient safety, and prescription accessibility simultaneously.

2. Objectives

- Read printed text from medicine strips using object detection and transformer-based OCR
- Recognise messy, cursive doctor handwriting from prescription photos
- Find cheaper generic alternatives by salt composition matching across 250,000+ Indian medicines
- Display top-3 side effects with frequency labels (Common / Uncommon / Rare)
- Warn users when two scanned medicines have dangerous drug interactions
- Detect fake or tampered packaging using unsupervised anomaly detection

3. System Requirements

Hardware: GPU with 4GB+ VRAM for model training; standard CPU for inference

Software: Python 3.10+, PyTorch, Django, OpenCV, HuggingFace Transformers, Ultralytics YOLOv8, pandas, SQLite

Data: 14 curated datasets from Kaggle, HuggingFace, Mendeley, EMBL, Columbia University, MVTec, and Roboflow

4. Deep Learning Models

Model	Purpose	Architecture	Datasets Used
Strip OCR	Reads brand name & dosage from medicine strip photo	YOLOv8 + TrOCR	Indian medicine strip images, phone-camera pharma packages, YOLO-annotated pill dataset
Prescription HTR	Converts doctor's handwritten prescription to clean text	CNN-BiLSTM-CTC	RxHandBD medical prescription words, illegible prescription images, HuggingFace prescription OCR dataset
Generic Finder	Finds cheaper medicines with same salt composition	CSV lookup + fuzzy match	A-Z India medicine database (250K), 1mg India dataset (300K, 2025), Indian medicine structured data
Side Effects + DDI	Shows top-3 side effects; warns on dangerous combinations	DB lookup + classifier	SIDER 4.1 (FDA side effects), TWOSIDES (Columbia Univ.), DrugBank interactions
Anomaly Detector	Flags tampered or counterfeit packaging	Autoencoder (unsupervised)	MVTec Anomaly Detection (pill & capsule), Counterfeit medicine detection dataset

5. What Makes MediScan Unique

- Five deep learning models in one integrated pipeline — most student projects use one
- Unsupervised anomaly detection trained only on real packaging; no fake images needed during training
- Fully camera-driven — zero typing, accessible to elderly and illiterate users across India
- Drug interaction data from Columbia University research (TWOSIDES) — used in Stanford and MIT papers